

Geographic Centroidal Voronoi Districts: Projected Proximity as a BISECT Split Primitive

T.11 — Algorithm Design / Geometric Structure Phase 2

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Abstract

Centroidal Voronoi Districts (CVD, T.10) now has two implemented metric modes inside BISECT: graph-distance CVD, which assigns tracts by BFS hop count, and geographic CVD, which assigns tracts by Euclidean distance after projecting tract centroids with the platform’s spherical Albers transform. This companion paper explains why the geographic mode exists, how it is wired into recursive bisection, and what still separates the implementation from a publication-grade state-by-state comparison.

The implemented changes over graph-distance CVD are concrete: a distinct geographic seed domain; k-farthest initialization in projected coordinate space; Lloyd-style coordinate updates snapped to the nearest tract; a one-meter seed-displacement stopping rule; and an error path when geographic CVD is requested without centroid data. The current implementation still rebalances each local split toward one half of node population, so ratio-aware geographic CVD splits for odd-seat nodes remain an implementation and evidence gap.

The paper therefore treats NC/FL/WA tables as benchmark targets, not final claims. Its main contribution is the audit contract: geographic CVD is now a BISECT split

primitive that can be verified from adjacency, population, centroid, base-seed, and node-path inputs.

Contents

1	Introduction	4
1.1	Why a Second CVD Metric Exists	4
1.2	Show the Difference	4
1.3	Implemented Scope	5
1.4	Relationship to T.10	5
1.5	Relation to Prior Work	5
2	Phase 1 vs. Phase 2: Property Comparison	6
2.1	Design Rationale for Each Difference	6
2.2	What BISECT Uses This For	7
3	Geographic CVD: Algorithm	8
3.1	Notation	8
3.2	Albers Equal-Area Conic Projection	8
3.3	Seed Derivation	9
3.4	K-Farthest Seed Initialisation (Euclidean)	10
3.5	Voronoi Assignment and Centroid Update	10
3.6	ε -Convergence Criterion	11
3.7	Formal Pseudocode	11
3.8	Show the Iteration	12
3.9	Convergence Analysis	13
3.10	Runtime Complexity	13
3.11	Audit Chain Fields	14
4	Benchmark Targets and Evidence Status	15
4.1	What the Current Evidence Proves	15
4.2	Publication Benchmark Plan	15
4.3	What to Compare	15

4.4	Failure Modes to Archive	16
5	Data Pipeline: Centroid Companion Files	17
5.1	File Format	17
5.2	Fetching Centroids	18
5.3	LoadedGraph Integration	19
5.4	Consistency with Analysis Tools	19
6	Conclusion	20

1. Introduction

1.1. Why a Second CVD Metric Exists

Centroidal Voronoi Districts (CVD, T.10 [Della-Libera, 2026]) are a seed-and-grow split primitive for BISECT. At each bisection node, the algorithm places two seeds, assigns every tract to its nearest seed, moves each seed to the center of the tract set it owns, and repeats. The important choice is what “nearest” means.

Graph-distance CVD answers nearest by adjacency hops. That is often a good redistricting default because contiguity is native to the graph: a tract one edge away is close in the bisection graph, and the medoid of a district is an actual tract. Geographic CVD answers nearest by projected coordinate distance: each tract has a longitude/latitude centroid, BISECT projects those centroids with its Albers transform, and Euclidean distance in meters decides ownership.

The two metrics see different worlds. A water body, mountain corridor, or long string of small urban tracts can make graph distance and map distance disagree. When they disagree, the bisection seed should not be chosen by accident; it should be chosen by an explicit metric that an auditor can reproduce.

1.2. Show the Difference

The following schematic shows the difference between saying “two tracts are close” in graph space and showing why geographic CVD may choose a different split. The letters are tracts, not districts. The gap is water or empty land: there is no adjacency edge across it, but the map distance across the gap is short.

Map view						Graph-hop view					
A	A	A		B	B	A-A-A-A-A-B-B-B-B-B land corridor only					
A	A			B	B						
A	A	A		B	B						

Graph-distance CVD grows through the available corridor. Geographic CVD sees the projected coordinates and may assign the right shore to the right seed even if the graph path between shores is long. This does not replace contiguity checking; it changes the proximity signal used before BISECT’s local rebalance and final validation.

1.3. Implemented Scope

This paper documents the implemented BISECT surface:

- Graph-distance CVD uses BFS hop count and graph-distance medoids.
- Geographic CVD uses projected tract centroids, Euclidean assignment, and population-weighted coordinate updates snapped to the nearest tract.
- "CVD_GEO_INIT_" domain-separates geographic CVD node seeds from graph-distance CVD seeds.
- Missing centroid data is a hard configuration error for geographic CVD, not a silent fallback to graph-distance CVD.

The current implementation does not yet provide state-specific projection dispatch, archived NC/FL/WA comparison packages, or ratio-aware CVD target populations for odd-seat bisection nodes. Those items are the evidence ceiling for this paper.

1.4. Relationship to T.10

T.10 is the canonical CVD paper: it explains proximity packing, the graph-distance implementation, and the shared recursive bisection contract. This T.11 note is the geographic-metric companion. It is narrower by design: where T.10 asks “what is CVD?”, this paper asks “what changes when CVD uses projected coordinate distance, and how does an auditor reproduce that choice?”

1.5. Relation to Prior Work

The continuous CVT framework of Du, Faber, and Gunzburger 1999 applies Lloyd’s algorithm in Euclidean space. Geographic CVD is a discrete, tract-level adaptation: the centroid is the population-weighted mean of tract coordinates, and the next seed is the real tract nearest to that mean. Snyder 1987 gives the Albers projection formulas used by the platform’s spherical transform; the present paper treats projection metadata as part of the audit package rather than as an incidental implementation detail.

Table 1: Phase 1 CVD vs. Phase 2 CVD-GEO: property comparison. The table describes the implemented contract, not a final empirical ranking.

Property	Phase 1 (GraphDistance)	Phase 2 (Geographic)
Distance metric	BFS hop-count on adjacency	Euclidean on projected (m)
Seed update	Graph-distance medoid (approximate, probe 50)	Pop-weighted mean coord (exact, nearest-snap)
Convergence criterion	Exact seed equality	$ \Delta_{\text{seed}} < 1.0 \text{ m}$
Data needed	Adjacency only	Adjacency + centroids
Centroid companion	Not required	<code>{stem}_centroids.json</code>
Seed prefix	<code>"CVD_INIT_"</code>	<code>"CVD_GEO_INIT_"</code>
CLI flag	<code>-cvd-metric graph-distance</code>	<code>-cvd-metric geographic</code>
BFS per medoid update	candidate BFS to probes	0 (arithmetic only)
Runtime per iteration	assignment BFS + medoid BFS	coordinate scans
Current split target	half-population rebalance	half-population rebalance
Best for	Regular-geometry inland states	Coastal, elongated, Alaska
Evidence status	L0/L1 covered; packages pending	L0/L1 covered; packages pending

2. Phase 1 vs. Phase 2: Property Comparison

Table 1 gives a direct property-by-property comparison of Phase 1 graph-distance CVD and Phase 2 geographic CVD-GEO. The rows span algorithmic, data, convergence, and implementation properties.

2.1. Design Rationale for Each Difference

Distance metric. The BFS hop-count metric requires no coordinate data and is correct for connected graphs with roughly uniform-area tracts. The Euclidean metric on Albers-projected coordinates requires centroid data but accurately represents geographic proximity regardless of tract size variation, coastal connectivity, or state shape. Phase 1 and Phase 2 are additive: the CLI defaults to graph-distance; geographic is an opt-in upgrade.

Centroid update. Phase 1 uses the approximate graph-distance medoid because the “centroid” of a cluster in hop-count space has no natural real-valued form: the mean of tract

indices is not a valid graph node. Phase 2 uses the population-weighted mean of projected (x, y) coordinates because the Euclidean centroid is well-defined and computable in $O(m)$ time. The nearest real tract to the computed centroid (by Euclidean distance) is used as the updated seed; this discrete rounding is the only approximation in Phase 2.

Convergence criterion. Phase 1 checks for exact seed equality between consecutive iterations: the same tract index must be the medoid in two successive rounds. Phase 2 checks that the Euclidean displacement of the projected seed is below $\varepsilon = 1.0$ m between iterations. The ε -criterion is useful because the nearest-tract snap can oscillate or make sub-meter coordinate moves: if two tracts are effectively equidistant from the computed centroid, exact coordinate equality is too brittle. A tolerance of 1 m resolves all practical oscillations at census-tract resolution (typical tract widths are 200–2,000 m).

Seed prefix. The Phase 2 prefix "CVD_GEO_INIT_" is distinct from the Phase 1 prefix "CVD_INIT_" by design. This ensures that the same `base_seed` and `node_path` produce different CVD node seeds for Phase 1 and Phase 2, preventing cross-phase seed collision in the audit chain. A platform invariant test asserts both constants are present in source and are not equal.

Data requirement and fallback behaviour. Geographic CVD requires a centroid companion file (`{stem}_centroids.json`) alongside the adjacency file. If the file is absent and `-cvd-metric geographic` is requested, the CLI returns an error naming centroid data and suggesting the centroid fetch path. The Phase 1 graph-distance path is unaffected: it requires no centroid data and remains the default.

2.2. What BISECT Uses This For

BISECT does not use geographic CVD to choose whether the statewide plan has k districts. That comes from apportionment and the bisection tree. It uses geographic CVD inside each internal tree node to choose the next two child regions. The input node already has a tract set; geographic CVD decides which tracts are closer to seed 0 and which are closer to seed 1, then BISECT carries those children forward.

Node before split	Geographic CVD ownership	Children																																				
<table><tr><td>.</td><td>.</td><td>.</td><td>.</td></tr><tr><td>.</td><td>.</td><td>.</td><td>.</td></tr><tr><td>.</td><td>.</td><td>.</td><td>.</td></tr></table>	.	.	.	.	.	.	.	.	.	.	.	.	<table><tr><td>0</td><td>0</td><td>1</td><td>1</td></tr><tr><td>0</td><td>0</td><td>1</td><td>1</td></tr><tr><td>0</td><td>1</td><td>1</td><td>1</td></tr></table>	0	0	1	1	0	0	1	1	0	1	1	1	<table><tr><td>L</td><td>L</td><td>R</td><td>R</td></tr><tr><td>L</td><td>L</td><td>R</td><td>R</td></tr><tr><td>L</td><td>R</td><td>R</td><td>R</td></tr></table>	L	L	R	R	L	L	R	R	L	R	R	R
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For odd-seat nodes, the bisection tree records child seat counts such as 3 : 4. The current CVD splitter still rebalances the local tract split toward one half of population. Ratio-aware CVD target populations are therefore a known implementation gap before geographic CVD should carry strong odd-seat publication claims.

3. Geographic CVD: Algorithm

3.1. Notation

Let $G_v = (V_v, E_v)$ be the subgraph at bisection tree node v , with $m = |V_v|$ tracts and population p_i for each tract i . Let $\ell_i = (\lambda_i, \phi_i)$ be the WGS84 (longitude, latitude) of the population-weighted centroid of tract i 's polygon, loaded from the centroid companion file (Section 5). Let $\mathbf{x}_i = (x_i, y_i) = \text{proj}(\ell_i)$ be the Albers-projected coordinates in meters (Section 3.2).

3.2. Albers Equal-Area Conic Projection

We use the EPSG:5070 spherical Albers formula from Snyder 1987, Chapter 14:

Definition 1 (Albers Projection — Continental US (EPSG:5070)). *Parameters: central meridian $\lambda_0 = -96^\circ$, standard parallels $\phi_1 = 29.5^\circ$, $\phi_2 = 45.5^\circ$. Given WGS84 (λ, ϕ) in*

degrees, convert to radians and compute:

$$n = \frac{1}{2}(\sin \phi_1 + \sin \phi_2), \quad (1)$$

$$C = \cos^2 \phi_1 + 2n \sin \phi_1, \quad (2)$$

$$\rho_0 = \sqrt{C}/n, \quad (3)$$

$$\rho = \sqrt{C - 2n \sin \phi}/n, \quad (4)$$

$$\theta = n(\lambda - \lambda_0), \quad (5)$$

$$x = R \cdot \rho \sin \theta, \quad (6)$$

$$y = R(\rho_0 - \rho \cos \theta), \quad (7)$$

where $R = 6,371,000$ m is the spherical Earth radius.

Remark 1 (Spherical approximation error). *Definition 1 uses a spherical Earth ($R = 6,371,000$ m). The EPSG:5070 standard uses the GRS80 ellipsoid; the difference in eccentricity ($e^2 = 0.00669438$) introduces at most 0.3% geometric error for continental US coordinates ($\lambda \in [-125^\circ, -66^\circ]$, $\phi \in [24^\circ, 50^\circ]$). At census-tract resolution (typical tract width 200–2,000 m), this corresponds to ≤ 3 m positional error per tract. For redistricting purposes, this error is negligible: it is far below the minimum meaningful districting unit (a census tract) and introduces no directional bias. An ellipsoidal implementation using PROJ library bindings is deferred to Phase 3.*

Remark 2 (Projection scope). *The current BISECT implementation applies the same spherical Albers transform for every geographic CVD run. State-specific projection dispatch for Alaska, Hawaii, Puerto Rico, and territories is a future evidence and implementation item. Any publication package using those jurisdictions should record the projection used and should not imply automatic EPSG dispatch until that code exists.*

3.3. Seed Derivation

Phase 2 uses a distinct seed prefix to isolate Phase 2 seeds from Phase 1 in the audit chain.

Definition 2 (Geographic CVD Node Seed). *Let `base_seed` be the unsigned 64-bit integer global seed and `node_path` the path string for bisection tree node v . The geographic CVD*

node seed is

$$s_v^{\text{geo}} = \text{first8}(\text{SHA-256}(\text{"CVD_GEO_INIT_"} \parallel \text{node_path} \parallel \text{"_"} \parallel \text{le8}(\text{base_seed}))),$$

where $\text{le8}(x)$ encodes x as 8 little-endian bytes and $\text{first8}(h)$ reads the first 8 bytes as a little-endian unsigned 64-bit integer [National Institute of Standards and Technology, 2015].

The prefix "CVD_GEO_INIT_" is strictly distinct from the Phase 1 prefix "CVD_INIT_": a test invariant asserts both constants are present in source and are not equal.

3.4. K-Farthest Seed Initialisation (Euclidean)

Definition 3 (K-Farthest Euclidean Initialisation). *Let $\text{sorted} = (t_0, \dots, t_{m-1})$ be tracts in V_v ordered by global index.*

1. **First seed:** $\text{seed}_0 = t_{s_v^{\text{geo}} \bmod m}$.
2. **Euclidean distances from $\mathbf{x}_{\text{seed}_0}$:** for each $t \in V_v$, $\delta(t) = \|\mathbf{x}_t - \mathbf{x}_{\text{seed}_0}\|$.
3. **Second seed:** $\text{seed}_1 = \arg \max_{t \in V_v} \delta(t)$ (Euclidean farthest; ties broken by sorted index).

The projected seed positions are $\mathbf{s}_0 = \mathbf{x}_{\text{seed}_0}$, $\mathbf{s}_1 = \mathbf{x}_{\text{seed}_1}$.

Unlike Phase 1 (BFS farthest), Definition 3 finds the two tracts that are geographically farthest apart in projected coordinates. This is an $O(m)$ scan rather than a BFS pass. For rectangular states, k-farthest Euclidean initialisation places seeds at the two geographic corners, immediately producing near-converged Voronoi regions.

3.5. Voronoi Assignment and Centroid Update

Definition 4 (Euclidean Voronoi Assignment). *Given projected seeds $(\mathbf{s}_0, \mathbf{s}_1)$, the assignment $a : V_v \rightarrow \{0, 1\}$ assigns each tract to its nearest seed by Euclidean distance:*

$$a(t) = \arg \min_{j \in \{0,1\}} \|\mathbf{x}_t - \mathbf{s}_j\|,$$

with ties broken in favour of district 0.

Definition 5 (Population-Weighted Centroid Update). *Let $D_j = \{t \in V_v : a(t) = j\}$ be the tracts in district j and $P_j = \sum_{t \in D_j} p_t$ the district population. The population-weighted centroid of D_j in projected space is*

$$\bar{\mathbf{c}}_j = \frac{1}{P_j} \sum_{t \in D_j} p_t \cdot \mathbf{x}_t.$$

The point $\bar{\mathbf{c}}_j$ may not correspond to any real tract. The updated seed position is the projected coordinate of the nearest real tract:

$$\mathbf{s}_j \leftarrow \mathbf{x}_{t^*}, \quad t^* = \arg \min_{t \in V_v} \|\mathbf{x}_t - \bar{\mathbf{c}}_j\|.$$

The nearest-tract snap (Definition 5) is the only approximation in Phase 2. In continuous CVT theory, the seed would be exactly at $\bar{\mathbf{c}}_j$; the snap replaces it with the geographically closest real tract. This approximation is exact when $\bar{\mathbf{c}}_j$ lies inside a real tract’s footprint and introduces at most one census-tract width of error otherwise.

3.6. ε -Convergence Criterion

Definition 6 (ε -Convergence). *At the end of each iteration, let $\mathbf{s}_j^{\text{new}}$ be the updated seed and $\mathbf{s}_j^{\text{prev}}$ the seed from the previous iteration. The algorithm terminates when:*

$$\max_{j \in \{0,1\}} \|\mathbf{s}_j^{\text{new}} - \mathbf{s}_j^{\text{prev}}\| < \varepsilon, \quad \varepsilon = 1.0 \text{ m}.$$

The 1 m tolerance is chosen to resolve discrete snap oscillations where two equidistant candidate tracts alternate as the nearest-tract to $\bar{\mathbf{c}}_j$. Typical census tract widths are ≥ 200 m; a convergence tolerance of 1 m is thus sub-tract-granularity and introduces no distributional bias.

3.7. Formal Pseudocode

Algorithm 1 gives the complete Phase 2 CVD-GEO bisection.

Algorithm 1 CVD-GEO-BISECT(G_v , $\mathbf{x}$, $\mathbf{p}$, base_seed , node_path , cvd_iters , balance_tolerance)

```

1:  $m \leftarrow |V_v|$ 
2: if  $m < 2$  then return degenerate assignment
3: end if
4:  $s_v \leftarrow \text{CVD-Geo-Seed}(\text{base\_seed}, \text{node\_path})$  ▷ Definition 2
5:  $\text{sorted} \leftarrow \text{sort}(V_v)$ ;  $\text{seed}_0 \leftarrow \text{sorted}[s_v \bmod m]$ 
6:  $\mathbf{s}_0 \leftarrow \mathbf{x}_{\text{seed}_0}$ ;  $\mathbf{s}_1 \leftarrow \mathbf{x}_{\arg \max_t \|\mathbf{x}_t - \mathbf{s}_0\|}$  ▷ k-farthest: Definition 3
7:  $\mathbf{s}_0^{\text{prev}} \leftarrow (-\infty, -\infty)$ ;  $\mathbf{s}_1^{\text{prev}} \leftarrow (-\infty, -\infty)$ 
8: for  $\text{iter} = 0$  to  $\text{cvd\_iters} - 1$  do
9:    $a(t) \leftarrow \arg \min_j \|\mathbf{x}_t - \mathbf{s}_j\|$  for all  $t \in V_v$  ▷ Voronoi: Definition 4
10:  for  $j \in \{0, 1\}$  do
11:     $D_j \leftarrow \{t : a(t) = j\}$ ;  $P_j \leftarrow \sum_{t \in D_j} p_t$ 
12:     $\bar{\mathbf{c}}_j \leftarrow P_j^{-1} \sum_{t \in D_j} p_t \cdot \mathbf{x}_t$  ▷ pop-weighted centroid: Definition 5
13:     $\mathbf{s}_j \leftarrow \mathbf{x}_{\arg \min_{t \in V_v} \|\mathbf{x}_t - \bar{\mathbf{c}}_j\|}$  ▷ nearest-tract snap
14:  end for
15:  if  $\max_j \|\mathbf{s}_j - \mathbf{s}_j^{\text{prev}}\| < 1.0$  then break ▷  $\varepsilon$ -convergence: Definition 6
16:  end if
17:   $\mathbf{s}_0^{\text{prev}} \leftarrow \mathbf{s}_0$ ;  $\mathbf{s}_1^{\text{prev}} \leftarrow \mathbf{s}_1$ 
18: end for
19:  $a \leftarrow \arg \min_j \|\mathbf{x}_t - \mathbf{s}_j\|$  for all  $t$  ▷ final assignment from converged seeds
20:  $(S, \bar{S}) \leftarrow \text{HalfRebalance}_{200}(a, G_v, \mathbf{p}, \text{balance\_tolerance})$  ▷ current implementation
    target
21: return  $(S, \bar{S})$ 

```

3.8. Show the Iteration

The geographic mode is easiest to audit as a sequence of ownership pictures. The first row shows two seeds in projected coordinate space. The second row shows Euclidean Voronoi ownership. The third row shows the population-weighted centroid of each owned side and the snap back to a real tract.

1. Seeds					2. Ownership					3. New seeds				
S0	.	.	.	S1	0	0	0	1	1	.	.	.	.	.
.	.	.	.	.	0	0	1	1	1	.	S0'	.	S1'	.
.	.	.	.	.	0	0	1	1	1	.	.	.	.	.

The point is not the labels “0” and “1”; it is the visible ownership frontier. Geographic CVD changes where that frontier falls because its distance field is drawn from projected coordinates instead of graph hops.

Key difference from Phase 1. The medoid computation (Phase 1: 100 BFS passes per iteration for the two probe sets) is replaced by a weighted mean plus a nearest-neighbour scan (Phase 2: $O(m)$ arithmetic per iteration with no BFS overhead). All other structural elements — k-farthest initialisation, post-hoc rebalance, audit chain output — are preserved.

3.9. Convergence Analysis

Proposition 1 (Bounded Lloyd Step for Geographic CVD). *Let $\mathcal{E}(\mathbf{s}_0, \mathbf{s}_1) = \sum_j \sum_{t \in D_j} \|\mathbf{x}_t - \mathbf{s}_j\|^2$ be the total Euclidean quantisation energy. The assignment and continuous centroid update are Lloyd steps in projected space. The nearest-tract snap adds a bounded discretisation error determined by the local tract-centroid spacing.*

Proof. (a) Voronoi assignment minimises $\mathcal{E}$ over all assignments given fixed seeds: assigning each tract to its nearest seed reduces $\sum_t \|\mathbf{x}_t - \mathbf{s}_{a(t)}\|^2$. (b) Population-weighted centroid update minimises $\mathcal{E}$ over seed positions given fixed assignment: the minimiser of $\sum_{t \in D_j} \|\mathbf{x}_t - \mathbf{c}\|^2$ over $\mathbf{c}$ is the population-weighted mean $\bar{\mathbf{c}}_j$. (c) Nearest-tract snap replaces the optimal $\bar{\mathbf{c}}_j$ with the closest real-tract projection $\mathbf{s}_j^* = \arg \min_t \|\mathbf{x}_t - \bar{\mathbf{c}}_j\|$. The additional energy from the snap is $\sum_{t \in D_j} (\|\mathbf{x}_t - \mathbf{s}_j^*\|^2 - \|\mathbf{x}_t - \bar{\mathbf{c}}_j\|^2) = 2(\mathbf{s}_j^* - \bar{\mathbf{c}}_j)^\top \sum_t (\mathbf{x}_t - \bar{\mathbf{c}}_j) + P_j \|\mathbf{s}_j^* - \bar{\mathbf{c}}_j\|^2$. The first term vanishes because $\bar{\mathbf{c}}_j$ is the centroid ($\sum_t p_t (\mathbf{x}_t - \bar{\mathbf{c}}_j) = 0$); the second term is $P_j \|\mathbf{s}_j^* - \bar{\mathbf{c}}_j\|^2$. The snap distance from the continuous centroid $\bar{\mathbf{c}}$ to the nearest real tract $c_* = \arg \min_{t \in D_j} \|p_t - \bar{\mathbf{c}}\|$ is bounded by the maximum nearest-neighbour radius $r_{\max} = \max_t \min_{t' \neq t} \|\mathbf{x}_t - \mathbf{x}_{t'}\|$ of the projected tract set. For census-tract graphs with typical inter-centroid spacing ≥ 500 m, and convergence threshold $\varepsilon = 1$ m, the snap displacement satisfies $\|c_* - \bar{\mathbf{c}}\| \leq r_{\max}$, which is bounded by the minimum tract size. Thus the total energy change per snap is $O(r_{\max}^2)$, bounded and converging. Combining (a), (b), and (c), the implemented update is a Lloyd-style descent step followed by bounded discrete snapping. Because the implementation also has a fixed `cvd_iters` cap, termination is guaranteed even when a discrete two-cycle would otherwise persist. $\square$

3.10. Runtime Complexity

Proposition 2 (Phase 2 Runtime). *CVD-GEO bisection at a node with m tracts requires:*

- *K-farthest initialisation: one Euclidean distance scan, $O(m)$.*
- *Per iteration:*

- Voronoi assignment: $O(m)$ (two Euclidean comparisons per tract).
- Centroid update: $O(m)$ weighted accumulation + $O(m)$ nearest scan.
- Post-hoc rebalance: $O(200m)$.
- Total: $O((n_{\text{iter}} + 200) \times m)$ for the implemented two-seed bisection node.

For $n_{\text{iter}} = 20$: $O(220m) = O(m)$. This removes the graph-medoid BFS passes used by graph-distance CVD. End-to-end speed still depends on centroid I/O, projection cost, rebalance work, and final validation, so the paper treats runtime comparison as an empirical benchmark target rather than a theorem.

3.11. Audit Chain Fields

The audit package for a geographic CVD run should expose, at minimum, the following fields or equivalent manifest entries. Some are implemented in logs or configuration today; others are acceptance targets for a complete archived package:

```
"structure": "centroidal-voronoi",
"cvd_metric": "geographic",           (was "graph-distance")
"n_iter": 20,
"seed_init": "kmeans++",
"base_seed": 12345678,
"cvd_seed": 987654321,
"cvd_seed_formula": "SHA-256('CVD_GEO_INIT_' || node_path || ...)",
"node_path": "01",                   (dagger)
"iterations_to_convergence": 11,
"convergence_warning": false,
"projection": "spherical Albers / EPSG:5070 constants", (dagger)
"final_ec": 2847,
"final_pp_mean": 0.241
```

The † fields are new in Phase 2. `cvd_seed_formula` is a literal string encoding the seed derivation so that an independent auditor can recompute `cvd_seed` from `base_seed` and `node_path` without consulting source code. `node_path` is required: the Phase 2 seed depends

on both `base_seed` and `node_path`, and without `node_path` the seed cannot be rederived from `base_seed` alone. `projection` should record the transform actually used. In the current code path, that is the platform spherical Albers transform with EPSG:5070 constants, not state-specific dispatch.

4. Benchmark Targets and Evidence Status

4.1. What the Current Evidence Proves

The implemented code proves that geographic CVD is a real BISECT split mode, not just a paper design. The L0/L1 surface covers:

- parsing `-cvd-metric geographic`;
- the distinct `"CVD_GEO_INIT_"` seed domain;
- successful `k=2` geographic CVD splits on synthetic centroid grids;
- deterministic output for the same base seed and node path;
- a hard error when geographic CVD is requested without centroid data; and
- the Albers central-meridian invariant used by the projection helper.

That evidence is sufficient for an implementation note. It is not sufficient for a final empirical claim such as “geographic CVD improves Florida by 6 percent.” Those claims require archived input packages, seed manifests, run directories, output hashes, and same-metric analysis artifacts.

4.2. Publication Benchmark Plan

Table 2 is therefore a benchmark target table. Its rows state what the publication package should measure once NC/FL/WA centroid packages are archived and reproducible. Values are intentionally marked TBD; a future evidence pass should replace them with run-derived numbers and cite the package hashes.

4.3. What to Compare

The benchmark should compare three effects separately.

Table 2: Phase 1 vs. geographic CVD benchmark matrix. Values are targets for an archived reproduction package, not current paper claims.

State	Algorithm	PP	EC	Pop dev.	Iters	Package
NC ($k = 14$)	Graph-distance CVD	TBD	TBD	TBD	TBD	pending
NC ($k = 14$)	Geographic CVD	TBD	TBD	TBD	TBD	pending
FL ($k = 28$)	Graph-distance CVD	TBD	TBD	TBD	TBD	pending
FL ($k = 28$)	Geographic CVD	TBD	TBD	TBD	TBD	pending
WA ($k = 10$)	Graph-distance CVD	TBD	TBD	TBD	TBD	pending
WA ($k = 10$)	Geographic CVD	TBD	TBD	TBD	TBD	pending

Metric effect. Run graph-distance CVD and geographic CVD with the same base seeds, the same bisection tree, the same population tolerance, and the same centroid package. Only the distance metric should differ.

Geometry effect. Use state families where graph distance and geographic distance are expected to diverge differently: a regular inland state such as North Carolina, a panhandle/coastal state such as Florida, and a water-barrier state such as Washington. The claim should be visual as well as numeric: show the seed frontier on at least one node where the two metrics split the tract grid differently.

Odd-seat effect. For $k = 7$, $k = 13$, $k = 27$, and similar odd-seat trees, record whether the local CVD split is being forced toward half-population rather than the tree’s $k_L : k_R$ child ratio. Until ratio-aware CVD splits are implemented, this effect should be disclosed beside any statewide compactness numbers.

4.4. Failure Modes to Archive

A useful geographic CVD package should include failures and boundary cases, not only successful runs:

- missing centroid companion file, expected configuration error;
- malformed centroid JSON, expected load error;
- centroid length mismatch, expected load error before split execution;

- synthetic grid with identical centroids, expected deterministic tie behavior; and
- real state with island or water-barrier geometry, expected visible metric divergence.

These cases are the bridge between an attractive geometric algorithm and an auditable election-administration tool: users need to know not just when the metric works, but how it fails.

5. Data Pipeline: Centroid Companion Files

5.1. File Format

Geographic CVD requires a centroid companion file alongside the adjacency file. The implemented loader derives the centroid filename from the adjacency binary stem. For `{stem}.adj.bin`, it looks for `{stem}_centroids.json` in the same directory:

```
data/2020/
north_carolina_adjacency_2020.adj.bin
north_carolina_adjacency_2020_geoids.json
north_carolina_adjacency_2020_centroids.json  <-- geographic CVD input
```

The JSON structure is a single array of `[longitude, latitude]` pairs in WGS84, one entry per census tract in the same index order as the adjacency file:

```
{
  "centroids": [
    [-78.8740, 35.2271],  // tract 0
    [-80.8431, 35.2270],  // tract 1
    ...                  // one entry per tract
  ]
}
```

Index alignment. The centroid file must have exactly m entries, where m is the number of tracts in the adjacency file. If `len(centroids)  $\neq$   $m$` , the loader returns an error:

```
ERROR: centroid file length mismatch: got 2659, expected 2660
      Check that centroids file was generated from the same
      adjacency file as the graph.
```

Centroid definition. The Rust loader treats each entry as a WGS84 tract centroid aligned to adjacency index order. The broader data package should record how those centroids were derived: TIGER internal points, geometric polygon centroids, or a population-weighted process. The algorithm consumes coordinate pairs; the publication package must document their provenance.

5.2. Fetching Centroids

The CLI error path suggests the following user-facing fetch route:

```
bisect fetch --type centroids --states NC --year 2020
```

This paper treats that command as the intended package workflow. A publication package should verify the command, archive the generated `_centroids.json` files, and record hashes beside the adjacency and GEOID files. To fetch centroids for all states once the fetch route is available:

```
bisect fetch --type centroids --year 2020 --workers 8
```

The fetch step should be idempotent: if the centroid file already exists and its length matches the adjacency file, the fetch can be skipped.

CLI error when centroids absent. If `-cvs-metric geographic` is specified and the centroid companion file is absent, the CLI returns:

```
ERROR: --cvs-metric geographic requires tract centroid data.
      Run: bisect fetch --type centroids
```

No fallback to Phase 1 occurs silently: if the user requests geographic metric and centroids are unavailable, the run fails with a clear diagnostic.

5.3. LoadedGraph Integration

The `LoadedGraph` struct gains one new field:

```
pub struct LoadedGraph {  
    // ... existing fields ...  
    /// WGS84 lon/lat tract centroid.  
    /// Empty if the companion file is absent.  
    pub tract_centroids: Vec<(f64, f64)>,  
}
```

Loading behaviour:

- After loading the adjacency file, derive the companion path from the adjacency binary stem and look for `_centroids.json`.
- If present and length-valid: load into `tract_centroids`.
- If absent: leave `tract_centroids` empty (no error at load time; error is deferred to Phase 2 dispatch if `-cvd-metric geographic` is requested).
- If present but length-mismatched: return error immediately at load time.

This design preserves full backward compatibility: all existing Phase 1 CVD runs, all METIS runs, and all other structure-layer algorithms are unaffected by the presence or absence of the centroid field.

5.4. Consistency with Analysis Tools

The Albers projection constants in `albers_project` (Definition 1) should be recorded beside the compactness analysis configuration. The current transform uses central meridian -96° , parallels $29.5^\circ/45.5^\circ$, and $R = 6,371,000$ m. A platform invariant test asserts that `proj(-96.0°, 37.5°)` returns $(|x| < 1,000 \text{ m}, y > 0)$: the central meridian projects to near-zero x (within 1 km) and positive y (positive northing in the projection).

6. Conclusion

Geographic CVD completes an important practical choice in the CVD family: BISECT can now split by graph-hop proximity or by projected-coordinate proximity. The graph-distance mode is data-light and native to contiguity. The geographic mode is useful when the map and the adjacency graph tell different proximity stories.

When to use geographic CVD. Use `-cvd-metric geographic` when the split should be driven by map distance: coastal states, peninsulas, long corridors, island-adjacent regions, and sparse tracts with very different physical sizes. Use the default `graph-distance` mode when centroid data is absent, when the graph topology is the intended proximity model, or when the publication package has not yet validated projection and centroid provenance.

Current command shape.

algorithm:

```
structure: centroidal-voronoi
weights: geographic
search: convergence
cvd_iters: 20
cvd_metric: geographic
balance_tolerance: 0.5
```

Legal and practitioner use. Geographic CVD should be presented as a reproducible compactness reference plan, not as a legal compliance certificate. Compactness standards vary by jurisdiction; Voting Rights Act analysis, communities of interest, incumbent handling, county and municipal splits, and state-specific criteria remain separate checks. The value of geographic CVD is that its proximity decision is auditable from centroids, projection constants, base seed, and node path.

Remaining work. The next paper-quality and implementation milestones are clear:

1. Build archived NC/FL/WA benchmark packages with graph-distance and geographic CVD side by side.

2. Add ratio-aware CVD target populations so odd-seat nodes can split by $k_L : k_R$ rather than only by half population.
3. Record projection metadata in the run/audit manifest.
4. Validate the centroid fetch route and archive centroid file hashes.
5. Add state-specific projection dispatch only after tests and package metadata can prove which transform was used.

Summary. T.10 explains CVD as a proximity-packing split primitive. T.11 makes the geographic metric legible: show the coordinate field, show the ownership frontier, show the snapped centroid update, and show the audit inputs needed to reproduce the split. That is the standard this family needs before we move from attractive geometry to durable public evidence.

References

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John P. Snyder. *Map Projections—A Working Manual*. USGS Professional Paper 1395. United States Geological Survey, 1987. Authoritative reference for Albers Equal-Area Conic projection (Chapter 14). Provides the spherical-Earth formulae used in T.11's `albers_project` function. Spherical approximation introduces $\leq 0.3\%$ maximum error for continental US coordinates vs. the full GRS80 ellipsoid; at census-tract resolution this is ≤ 3 m and negligible for redistricting.